13.3 HIGHER-ORDER APPROACHES

More refined linearization techniques can be used to reduce the linearization error in the EKF for highly nonlinear systems. In this section, we will derive and illustrate two such approaches: the iterated EKF, and the second-order EKF. We will also briefly discuss other approaches, including Gaussian sum filters and grid filters.

13.3.1 The iterated extended Kalman filter

In this section, we will discuss the iterated EKF. We will confine our discussion here to discrete-time filtering, although the concepts can easily be extended to continuous or hybrid filters.

When we derived the discrete-time EKF in Section 13.2.3, we approximated $h(x_k, v_k)$ by expanding it in a Taylor series around $\hat{x}_k^-$, as shown in Equation (13.41):

$$\begin{aligned} h(x_k, v_k) &= h(\hat{x}_k^-, 0) + \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^-} (x_k - \hat{x}_k^-) + \left.\frac{\partial h}{\partial v}\right|_{\hat{x}_k^-} v_k \\ &= h(\hat{x}_k^-, 0) + H_k(x_k - \hat{x}_k^-) + M_k v_k \end{aligned} \tag{13.50}$$

Based on this linearization, we then wrote the measurement-update equations as shown in Equation (13.43):

$$\begin{aligned} K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + M_k R_k M_k^T)^{-1} \\ P_k^+ &= (I - K_k H_k) P_k^- \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k [y_k - h_k(\hat{x}_k^-, 0)] \end{aligned} \tag{13.51}$$

The reason that we expanded $h(x_k)$ around $\hat{x}_k^-$ was because that was our best estimate of x_k before the measurement at time k is taken into account. But after we implement the discrete EKF equations to obtain the *a posteriori* estimate $\hat{x}_k^+$, we have a better estimate of x_k. So we can reduce the linearization error by reformulating the Taylor series expansion of $h(x_k)$ around our new estimate. If we then use that new Taylor series expansion of $h(x_k)$ and recalculate the measurement-update equations, we should get a better a posteriori estimate of $\hat{x}_k^+$. But then we can repeat the previous step; since we have an even better estimate of x_k, we can again reformulate the expansion of $h(x_k)$ around this even better estimate to get an even *better* estimate. This process can be repeated as many times as desired, although for most problems the majority of the possible improvement is obtained by only relinearizing one time.

We use the notation $\hat{x}_{k,i}^+$ to refer to the *a posteriori* estimate of x_k after i relinearizations have been performed. So $\hat{x}_{k,0}$ is the *a posteriori* estimate that results from the application of the standard EKF. Likewise, we use $P_{k,i}^+$ to refer to the approximate estimation-error covariance of $\hat{x}_{k,i}^+$, $K_{k,i}$ to refer to the Kalman gain that is used during the ith relinearization step, and $H_{k,i}$ to refer to the partial derivative matrix evaluated at the $x_k = \hat{x}_{k,i}^+$.

With this notation, we can describe an algorithm for the iterated EKF as follows. First, at each time step k we initialize the iterated EKF estimate to the standard EKF estimate:

$$\begin{aligned} \hat{x}_{k,0}^+ &= \hat{x}_k^+ \\ P_{k,0}^+ &= P_k^+ \end{aligned} \tag{13.52}$$

Second, for $i = 0, 1, \cdots, N$, evaluate the following equations:

$$\begin{aligned} H_{k,i} &= \left.\frac{\partial h}{\partial x}\right|_{\hat{x}^+_{k,i}} \\ K_{k,i} &= P^-_k H^T_{k,i}(H_{k,i}P^-_k H^T_{k,i} + M_k R_k M^T_k)^{-1} \\ P^+_{k,i+1} &= (I - K_{k,i}H_{k,i})P^-_k \\ \hat{x}^+_{k,i+1} &= \hat{x}^-_k + K_{k,i}[y_k - h_k(\hat{x}^-_k)] \end{aligned} \tag{13.53}$$

This is done for as many steps as desired to improve the linearization. If $N = 0$ then the iterated EKF reduces to the standard EKF.

We still have to make one more modification to the above equations to obtain the iterated Kalman filter. Recall that in the derivation of the EKF, the $\hat{x}$ measurement update equation was originally derived from the following first-order Taylor series expansion of the measurement equation:

$$\begin{aligned} y_k &= h(x_k) \\ &\approx h(\hat{x}^-_k) + H|_{\hat{x}^-_k}(x_k - \hat{x}^-_k) \end{aligned} \tag{13.54}$$

To derive the measurement-update equation for $\hat{x}$ we evaluated the right side at the *a priori* estimate $\hat{x}^-_k$ and subtracted from y_k to get our correction term (the residual):

$$\begin{aligned} r_k &= y_k - h(\hat{x}^-_k) - H|_{\hat{x}^-_k}(\hat{x}^-_k - \hat{x}^-_k) \\ &= y_k - h(\hat{x}^-_k) \end{aligned} \tag{13.55}$$

With the iterated EKF we instead want to expand the measurement equation around $\hat{x}^+_{k,i}$ as follows:

$$y_k \approx h(\hat{x}^+_{k,i}) + H|_{\hat{x}^+_{k,i}}(x_k - \hat{x}^+_{k,i}) \tag{13.56}$$

To derive the iterated EKF measurement-update equation for $\hat{x}$, we evaluate the right side of the above equation at the *a priori* estimate $\hat{x}^-_k$ and subtract from y_k to get our correction term:

$$r_k = y_k - h(\hat{x}^+_{k,i}) - H_{k,i}(\hat{x}^-_k - \hat{x}^+_{k,i}) \tag{13.57}$$

This gives the iterated EKF update equation for $\hat{x}$ as

$$\hat{x}^+_{k,i+1} = \hat{x}^-_k + K_{k,i}[y_k - h(\hat{x}^+_{k,i}) - H_{k,i}(\hat{x}^-_k - \hat{x}^+_{k,i})] \tag{13.58}$$

The iterated EKF can then be summarized as follows.

The iterated extended Kalman filter

1. The nonlinear system and measurement equations are given as follows:

$$\begin{aligned} x_k &= f_{k-1}(x_{k-1}, u_{k-1}, w_{k-1}) \\ y_k &= h_k(x_k, v_k) \\ w_k &\sim (0, Q_k) \\ v_k &\sim (0, R_k) \end{aligned} \tag{13.59}$$

2. Initialize the filter as follows.

$$\begin{aligned} \hat{x}_0^+ &= E(x_0) \\ P_0^+ &= E\left[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T\right] \end{aligned} \tag{13.60}$$

3. For $k = 1, 2, \cdots$, do the following.

   (a) Perform the following time-update equations:

$$\begin{aligned} P_k^- &= F_{k-1} P_{k-1}^+ F_{k-1}^T + L_{k-1} Q_{k-1} L_{k-1}^T \\ \hat{x}_k^- &= f_{k-1}(\hat{x}_{k-1}^+, u_{k-1}, 0) \end{aligned} \tag{13.61}$$

   where the partial derivative matrices F_{k-1} and L_{k-1} are defined as follows:

$$\begin{aligned} F_{k-1} &= \left.\frac{\partial f_{k-1}}{\partial x}\right|_{\hat{x}_{k-1}^+} \\ L_{k-1} &= \left.\frac{\partial f_{k-1}}{\partial w}\right|_{\hat{x}_{k-1}^+} \end{aligned} \tag{13.62}$$

   Up to this point the iterated EKF is the same as the standard discrete-time EKF.

   (b) Perform the measurement update by initializing the iterated EKF estimate to the standard EKF estimate:

$$\begin{aligned} \hat{x}_{k,0}^+ &= \hat{x}_k^- \\ P_{k,0}^+ &= P_k^- \end{aligned} \tag{13.63}$$

   For $i = 0, 1, \cdots, N$, evaluate the following equations (where N is the desired number of measurement-update iterations):

$$\begin{aligned} H_{k,i} &= \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_{k,i}^+} \\ M_{k,i} &= \left.\frac{\partial h}{\partial v}\right|_{\hat{x}_{k,i}^+} \\ K_{k,i} &= P_k^- H_{k,i}^T \left(H_{k,i} P_k^- H_{k,i}^T + M_{k,i} R_k M_{k,i}^T\right)^{-1} \\ P_{k,i+1}^+ &= (I - K_{k,i} H_{k,i}) P_k^- \\ \hat{x}_{k,i+1}^+ &= \hat{x}_k^- + K_{k,i}\left[y_k - h(\hat{x}_{k,i}^+) - H_{k,i}(\hat{x}_k^- - \hat{x}_{k,i}^+)\right] \end{aligned} \tag{13.64}$$

   (c) The final *a posteriori* state estimate and estimation-error covariance are given as follows:

$$\begin{aligned} \hat{x}_k^+ &= \hat{x}_{k,N+1}^+ \\ P_k^+ &= P_{k,N+1}^+ \end{aligned} \tag{13.65}$$

An illustration of the iterated EKF will be presented in Example 13.3.

13.3.2 The second-order extended Kalman filter

The second-order EKF is similar to the iterated EKF in that it attempts to reduce the linearization error of the EKF. In the iterated EKF of the previous section, we refined the point at which we performed a first-order Taylor series expansion of the measurement equation $h(\cdot)$. In the second-order EKF we instead perform a *second-order* Taylor series expansion of $f(\cdot)$ and $h(\cdot)$. The second-order EKF presented in this section is based on [Ath68, Gel74].

In this section, we will consider the hybrid system with continuous-time system dynamics and discrete-time measurements:

$$\begin{aligned}
\dot{x} &= f(x,u,w,t) \\
y_k &= h(x_k,t_k) + v_k \\
w(t) &\sim (0,Q) \\
v_k &\sim (0,R_k)
\end{aligned} \tag{13.66}$$

In the standard EKF, we expanded $f(x,u,w,t)$ using a first-order Taylor series. In this section, we will consider only the expansion around a nominal x, ignoring the expansion around nominal u and w values. This is done so that we can present the main ideas of the second-order EKF without getting too bogged down in notation. The development in this section can be easily extended to second-order expansions around u and w once the main idea is understood.

The first-order expansion of $f(x,u,w,t)$ around $x = \hat{x}$ is given as

$$f(x,u,w,t) = f(\hat{x},u_0,w_0,t) + \left.\frac{\partial f}{\partial x}\right|_{\hat{x}} (x-\hat{x}) \tag{13.67}$$

In the standard EKF, we evaluated this expression at $x = \hat{x}$ to obtain our time-update equation for $\hat{x}$ as

$$\dot{\hat{x}} = f(\hat{x},u_0,w_0,t) \tag{13.68}$$

In the second-order EKF we expand $f(x,u,w,t)$ with an additional term in the Taylor series:

$$f(x,u,w,t) = f(\hat{x},u_0,w_0,t) + \left.\frac{\partial f}{\partial x}\right|_{\hat{x}} (x-\hat{x}) + \frac{1}{2}\sum_{i=1}^{n} \phi_i (x-\hat{x})^T \left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}} (x-\hat{x}) \tag{13.69}$$

where n is the dimension of the state vector, f_i is the ith element of $f(x,u,w,t)$, and the ϕ_i vector is defined as an $n \times 1$ vector with all zeros except for a one in the ith element. The quadratic term in the summation can be written as

$$(x-\hat{x})^T \left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}} (x-\hat{x}) = \operatorname{Tr}\left[\left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}} (x-\hat{x})(x-\hat{x})^T \right] \tag{13.70}$$

Since we do not know the value of $(x-\hat{x})(x-\hat{x})^T$ in the above equation, we replace it with its expected value, which is the covariance of the Kalman filter, to obtain

$$(x-\hat{x})^T \left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}} (x-\hat{x}) \approx \operatorname{Tr}\left[\left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}} P \right] \tag{13.71}$$

We then evaluate Equation (13.69) at $x = \hat{x}$ and substitute the above expression in the summation to obtain the time-update equation for $\hat{x}$ as

$$\dot{\hat{x}} = f(\hat{x}, u_0, w_0, t) + \frac{1}{2}\sum_{i=1}^{n} \phi_i \mathrm{Tr}\left[\left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}} P \right] \tag{13.72}$$

The time-update equation for P remains the same as in the standard hybrid EKF as shown in Equation (13.28):

$$\dot{P} = FP + PF^T + LQL^T \tag{13.73}$$

Now we will derive the measurement-update equations. Suppose that the measurement-update equation for the state estimate is given as

$$\hat{x}_k^+ = \hat{x}_k^- + K_k \left[y_k - h(\hat{x}_k^-, t_k) \right] - \pi_k \tag{13.74}$$

where K_k is the Kalman gain to be determined, and π_k is a correction term to be determined. We will choose π_k so that the estimate $\hat{x}_k^+$ is unbiased, and we will then choose K_k to minimize the trace of the covariance of the estimate.

If we define the estimation errors as

$$\begin{aligned} e_k^- &= x_k - \hat{x}_k^- \\ e_k^+ &= x_k - \hat{x}_k^+ \end{aligned} \tag{13.75}$$

we can see from Equations (13.66) and (13.74) that

$$e_k^+ = e_k^- - K_k \left[h(x_k, t_k) - h(\hat{x}_k^-, t_k) \right] - K_k v_k + \pi_k \tag{13.76}$$

Now we perform a second-order Taylor series expansion of $h(x_k, t_k)$ around the nominal point $\hat{x}_k^-$ to obtain

$$\begin{aligned} h(x_k, t_k) &= h(\hat{x}_k^-, t_k) + \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^-} (x_k - \hat{x}_k^-) + \\ &\quad \frac{1}{2}\sum_{i=1}^{m} \phi_i (x_k - \hat{x}_k^-)^T \left.\frac{\partial^2 h(i)}{\partial x^2}\right|_{\hat{x}_k^-} (x_k - \hat{x}_k^-) \\ &= h(\hat{x}_k^-, t_k) + H_k (x_k - \hat{x}_k^-) + \frac{1}{2}\sum_{i=1}^{m} \phi_i (x_k - \hat{x}_k^-)^T \left.\frac{\partial^2 h_i}{\partial x^2}\right|_{\hat{x}_k^-} (x_k - \hat{x}_k^-) \end{aligned} \tag{13.77}$$

where H_k is defined by the above equation, m is the dimension of the measurement vector, and h_i is the ith element of $h(x_k, t_k)$. This gives the *a posteriori* estimation error as

$$e_k^+ = e_k^- - K_k H_k e_k^- - \frac{1}{2} K_k \sum_{i=1}^{m} \phi_i (e_k^-)^T D_{k,i} e_k^- - K_k v_k + \pi_k \tag{13.78}$$

where $D_{k,i}$ is defined as

$$D_{k,i} = \left.\frac{\partial^2 h_i}{\partial x^2}\right|_{\hat{x}_k^-} \tag{13.79}$$

Taking the expected value of both sides of Equation (13.78), assuming that $E(e_k^-) = 0$, and making the same approximation as in Equation (13.71), we can see that in order to have $E(e_k^+) = 0$ we must set

$$\pi_k = \frac{1}{2} K_k \sum_{i=1}^{m} \phi_i \mathrm{Tr}\left[D_{k,i} P_k^-\right] \tag{13.80}$$

Defining P_k^+ as

$$P_k^+ = E\left[e_k^+ (e_k^+)^T\right] \tag{13.81}$$

and using the above equations, it can be shown after some involved algebraic calculations [Ath68] that

$$P_k^+ = (I - K_k H_k) P_k^- (I - K_k H_k)^T + K_k (R_k + \Lambda_k) K_k^T \tag{13.82}$$

where the matrix Λ_k is defined as

$$\Lambda_k = \frac{1}{4} E\left\{ \left[\sum_{i=1}^{m} \phi_i \mathrm{Tr}[D_{k,i}(e_k^- (e_k^-)^T - P_k^-)] \right] \left[\cdots \right]^T \right\} \tag{13.83}$$

Now we define a cost function J_k that we want to minimize as a weighted sum of estimation errors:

$$\begin{aligned} J_k &= E[(e_k^+)^T S_k e_k^+] \\ &= \mathrm{Tr}[S_k P_k^+] \end{aligned} \tag{13.84}$$

where S_k is any positive definition weighting matrix. The K_k that minimizes this cost function can be found as

$$K_k = P_k^- H_k^T \left(H_k P_k^- H_k^T + R_k + \Lambda_k \right)^{-1} \tag{13.85}$$

This gives the P_k^+ matrix from Equation (13.82) as

$$P_k^+ = P_k^- - P_k^- H_k^T \left(H_k P_k^- H_k^T + R_k + \Lambda_k \right)^{-1} H_k P_k^- \tag{13.86}$$

Now we need to figure out how to evaluate the Λ_k matrix in Equation (13.83). Note that Λ_k can be written as the double summation

$$\Lambda_k = \frac{1}{4} E\left\{ \sum_{i,j=1}^{m} \phi_i \phi_j^T \mathrm{Tr}\left[D_{k,i}(e_k^- (e_k^-)^T - P_k^-)\right] \mathrm{Tr}\left[D_{k,j}(e_k^- (e_k^-)^T - P_k^-)\right] \right\} \tag{13.87}$$

The product $\phi_i \phi_j^T$ is an $m \times m$ matrix whose elements are all zero except for the element in the ith row and jth column. Therefore, the element in the ith row and jth column of Λ_k can be written as

$$\Lambda_k(i,j) = \frac{1}{4} E\left\{ \mathrm{Tr}\left[D_{k,i}(e_k^- (e_k^-)^T - P_k^-)\right] \mathrm{Tr}\left[D_{k,j}(e_k^- (e_k^-)^T - P_k^-)\right] \right\} \tag{13.88}$$

This expression can be evaluated with the following lemma [Ath68].

Lemma 6 *Suppose we have the n-element random vector $x \sim N(0, P)$. Then*

$$\begin{aligned} E\left[x\, Tr(Axx^T)\right] &= 0 \\ E\left[Tr(Axx^TBxx^T)\right] &= E\left[Tr(Axx^T)\,Tr(Bxx^T)\right] \\ &= 2\,Tr(APBP) + Tr(AP)\,Tr(BP) \end{aligned} \tag{13.89}$$

where A and B are arbitrary $n \times n$ matrices.

Using this lemma with Equation (13.88) we can see that

$$\Lambda_k(i,j) = \frac{1}{2}\mathrm{Tr}(D_{k,i}P_k^- D_{k,j}P_k^-) \tag{13.90}$$

This equation, along with Equations (13.74), (13.80), (13.82), and (13.85), specify the measurement-update equations for the second-order EKF. The second-order EKF can be summarized as follows.

The second-order hybrid extended Kalman filter

1. The system equations are given as follows:

$$\begin{aligned} \dot{x} &= f(x, u, w, t) \\ y_k &= h(x_k, t_k) + v_k \\ w(t) &\sim (0, Q) \\ v_k &\sim (0, R_k) \end{aligned} \tag{13.91}$$

2. The estimator is initialized as follows:

$$\begin{aligned} \hat{x}_0^+ &= E(x_0) \\ P_0^+ &= E\left[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T\right] \end{aligned} \tag{13.92}$$

3. The time-update equations are given as

$$\begin{aligned} \dot{\hat{x}} &= f(\hat{x}, u, 0, t) + \frac{1}{2}\sum_{i=1}^{n} \phi_i \mathrm{Tr}\left[\left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}} P \right] \\ \dot{P} &= FP + PF^T + LQL^T \\ \phi_i &= \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{array}{l} \\ \\ \\ \leftarrow i\text{th element} \\ \\ \\ \\ \end{array} \\ F &= \left.\frac{\partial f}{\partial x}\right|_{\hat{x}} \\ L &= \left.\frac{\partial f}{\partial w}\right|_{\hat{x}} \end{aligned} \tag{13.93}$$

4. The measurement update equations are given as

$$
\begin{aligned}
\hat{x}_k^+ &= \hat{x}_k^- + K_k\left[y_k - h(\hat{x}_k^-)\right] - \pi_k \\
\pi_k &= \frac{1}{2}K_k\sum_{i=1}^{m}\phi_i \mathrm{Tr}\left[D_{k,i}P_k^-\right] \\
D_{k,i} &= \left.\frac{\partial^2 h_i(x_k,t_k)}{\partial x^2}\right|_{\hat{x}_k^-} \\
K_k &= P_k^- H_k^T\left(H_k P_k^- H_k^T + R_k + \Lambda_k\right)^{-1} \\
H_k &= \left.\frac{\partial h(x_k,t_k)}{\partial x}\right|_{\hat{x}_k^-} \\
\Lambda_k(i,j) &= \frac{1}{2}\mathrm{Tr}(D_{k,i}P_k^- D_{k,j}P_k^-) \\
P_k^+ &= P_k^- - P_k^- H_k^T\left(H_k P_k^- H_k^T + R_k + \Lambda_k\right)^{-1} H_k P_k^- \qquad (13.94)
\end{aligned}
$$

Note that setting the second partial derivative matrices in this algorithm to zero matrices results in the standard hybrid EKF.

■ **EXAMPLE 13.3**

In this example, we compare the performance of the EKF, the second-order EKF, and the iterated EKF for the falling body problem described in Example 13.2. A similar comparison was shown in [Wis69], where it was concluded that the iterated EKF had better RMS error performance, but the second-order filter had smaller bias. The system equations are the same as those shown in Example 13.2:

$$
\begin{aligned}
\dot{x}_1 &= x_2 + w_1 \\
\dot{x}_2 &= \rho_0 \exp(-x_1/k)x_2^2 x_3/2 - g + w_2 \\
\dot{x}_3 &= w_3 \qquad (13.95)
\end{aligned}
$$

In this example, we change the measurement system so that it does not measure the altitude of the falling body, but instead measures the range to the measuring device. The measuring device is located at an altitude a and at a horizontal distance M from the body's vertical line of fall. The measurement equation is therefore given by

$$
\begin{aligned}
y_k &= \sqrt{M^2 + (x_1(t_k) - a)^2} + v_k \\
&= h(x_k) + v_k \qquad (13.96)
\end{aligned}
$$

This makes the problem more nonlinear and hence more difficult to estimate (i.e., in Example 13.2 we had a nonlinear system but a linear measurement, whereas in this example we have nonlinearities in both the system and the measurement equations). The partial derivative F matrix for the EKFs are given in Example 13.2. The other partial derivative matrices used in the second-order EKF are given as follows:

$$
\begin{aligned}
H &= \frac{\partial h}{\partial x} \\
&= \left[\begin{array}{ccc} (x_1 - a)(M^2 + (x_1 - a)^2)^{-1/2} & 0 & 0 \end{array}\right] \\
L &= \frac{\partial f}{\partial w} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
D_1 &= \frac{\partial^2 h_1}{\partial x^2} \\
&= \begin{bmatrix} h^{-1}(1 - (x_1 - a)^2 h^{-2}) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
\frac{\partial^2 f_1}{\partial x^2} = \frac{\partial^2 f_3}{\partial x^2} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
\frac{\partial^2 f_2}{\partial x^2} &= \rho_0 \exp(-x_1/k) \begin{bmatrix} x_2^2 x_3/2k^2 & -x_2 x_3/k & -x_2^2/2k \\ -x_2 x_3/k & x_3 & x_2 \\ -x_2^2/2k & x_2 & 0 \end{bmatrix}
\end{aligned}
\tag{13.97}
$$

Table 13.2 shows the performances of the EKFs (averaged over 20 simulation runs). It is seen that second-order EKF provides significant improvement over the first-order EKF for altitude and velocity estimation, but for some reason it actually provides worse performance for ballistic coefficient estimation. Also note that the iterated EKF provides only slight improvement over the first-order EKF, and (as expected) the iterated EKF performs better when more iterations are executed for the linearization refinement.

Table 13.2 Example 13.3 results. A comparison of the estimation errors of different EKF approaches for tracking a falling body.

Filter	Altitude	Velocity	Ballistic Coefficient
First-order EKF	758 feet	518 feet/sec	0.091 feet^3/lb/sec^2
Second-order EKF	356	483	0.129
Iterated EKF ($N = 2$)	755	517	0.091
Iterated EKF ($N = 3$)	745	516	0.091
Iterated EKF ($N = 4$)	738	509	0.091
Iterated EKF ($N = 5$)	733	506	0.091
Iterated EKF ($N = 6$)	723	506	0.091

We conclude from this that the second-order filter has better estimation performance. However, the implementation is much more difficult and requires the computation of second-order derivatives. In this example, the second-order derivatives could be taken analytically because we have explicit analytical system and measurement equations. In many applications second-order derivatives will not be available analytically, and approximations will inevitably be subject to error.

These results are different than reported in [Wis69], where it was shown that the iterated EKF performed better than the second-order EKF. The different conclusions between this book and [Wis69] show that comparisons between different algorithms are often subjective. Perhaps the discrepancies are due to differences in implementations of the filtering algorithms, differences in implementations of the system dynamics or random noise generation, differences in the way that the estimation errors were measured, or even differences in the computing platforms that were used.

$\nabla\nabla\nabla$

The second-order filter was initially developed by Bass [Bas66] and Jazwinski [Jaz66]. A Gaussian second-order filter was developed by Athans [Ath68] and Jazwinski [Jaz70], in which fourth-order terms in Taylor series approximations are retained and approximated by assuming that the underlying probabilities are Gaussian. A small correction in the original derivations of the second-order EKF was reported by Rolf Henriksen [Hen82]. Although the second-order filter often provides improved performance over the extended Kalman filter, nothing definitive can be said about its performance, as evidenced by an example of an unstable second-order filter reported in [Kus67]. Additional comparison and analysis of some nonlinear Kalman filters can be found in [Sch68, Wis69, Wis70, Net78]. A simplified version of Henriksen's discrete-time second-order filter can be summarized as follows.

The second-order discrete-time extended Kalman filter

1. The system equations are given as follows:

$$\begin{aligned} x_{k+1} &= f(x_k, u_k, k) + w_k \\ y_k &= h(x_k, k) + v_k \\ w_k &\sim (0, Q_k) \\ v_k &\sim (0, R_k) \end{aligned} \tag{13.98}$$

2. The estimator is initialized as follows:

$$\begin{aligned} \hat{x}_0^+ &= E(x_0) \\ P_0^+ &= E\left[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T\right] \end{aligned} \tag{13.99}$$

3. The time update equations are given as follows:

$$\begin{aligned} \hat{x}_{k+1}^- &= f(\hat{x}_k^+, u_k, k) + \frac{1}{2}\sum_{i=1}^{n} \phi_i \mathrm{Tr}\left[\left.\frac{\partial^2 f_i}{\partial x}\right|_{\hat{x}_k^+} P_k^+\right] \\ P_{k+1}^- &= F P_k^+ F^T + Q_k \end{aligned}$$

$$
\begin{aligned}
\phi_i &= \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} \\ \\ \\ \leftarrow i\text{th element} \\ \\ \\ \\ \end{matrix} \\
F &= \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_k^+}
\end{aligned} \tag{13.100}
$$

4. The measurement update equations are given as follows:

$$
\begin{aligned}
\hat{x}_k^+ &= \hat{x}_k^- + K_k[y_k - h(\hat{x}_k^-, k)] - \pi_k \\
\pi_k &= \frac{1}{2} K_k \sum_{i=1}^{m} \phi_i \mathrm{Tr}\left[D_{k,i} P_k^-\right] \\
D_{k,i} &= \left.\frac{\partial^2 h_i(x_k, k)}{\partial x^2}\right|_{\hat{x}_k^-} \\
K_k &= P_k^- H_k^T \left(H_k P_k^- H_k^T + R_k\right)^{-1} \\
H_k &= \left.\frac{\partial h(x_k, k)}{\partial x}\right|_{\hat{x}_k^-} \\
P_k^+ &= (I - K_k H_k) P_k^-
\end{aligned} \tag{13.101}
$$

A more general version of the above algorithm can be found in [Hen82]. Similar to the hybrid second-order EKF presented earlier in this section, we note that setting the second-order partial derivative matrices in this algorithm to zero matrices results in the standard discrete-time EKF.

13.3.3 Other approaches

We have considered a couple of higher-order approaches to reducing the linearization error of the EKF. We looked at the iterated EKF and the second-order EKF, but other approaches are also available. For example, Gaussian sum filters are based on the idea that a non-Gaussian pdf can be approximated by a sum of Gaussian pdfs. This is similar to the idea that any curve can be approximated by a piecewise constant function. Since the true pdf of the process noise and measurement noise can be approximated by a sum of M Gaussian pdfs, we can run M Kalman filters in parallel on M Gaussian filtering problems, each of them optimal filters, and then combine them to obtain an approximately optimal estimate. The number of filters M is a trade-off between approximation accuracy (and hence optimality) and computational effort. This idea was first mentioned in [Aok65] and was explored in [Cam68, Sor71b, Als74, Kit89]. The Gaussian sum filter algorithm presented in [Als72] can be summarized as follows.

The Gaussian sum filter

1. The discrete-time n-state system and measurement equations are given as follows:

$$\begin{aligned} x_k &= f_{k-1}(x_{k-1}, u_{k-1}, w_{k-1}) \\ y_k &= h_k(x_k, v_k) \\ w_k &\sim (0, Q_k) \\ v_k &\sim (0, R_k) \end{aligned} \tag{13.102}$$

2. Initialize the filter by approximating the pdf of the initial state as follows:

$$\text{pdf}(\hat{x}_0^+) = \sum_{i=1}^{M} a_{0i} N(\hat{x}_{0i}^+, P_{0i}^+) \tag{13.103}$$

The a_{0i} coefficients (which are positive and add up to 1), the $\hat{x}_{0i}^+$ means, and the P_{0i}^+ covariances, are chosen by the user to provide a good approximation to the pdf of the initial state.

3. For $k = 1, 2, \cdots$, do the following.

   (a) The *a priori* state estimate is obtained by first executing the following time-update equations for $i = 1, \cdots, M$:

$$\begin{aligned} \hat{x}_{ki}^- &= f_{k-1}(\hat{x}_{k-1,i}^+, u_{k-1}, 0) \\ F_{k-1,i} &= \left.\frac{\partial f_{k-1}}{\partial x_{k-1}}\right|_{\hat{x}_{k-1,i}^+} \\ P_{ki}^- &= F_{k-1,i} P_{k-1,i}^+ F_{k-1,i}^T + Q_{k-1} \\ a_{ki} &= a_{k-1,i} \end{aligned} \tag{13.104}$$

   The pdf of the *a priori* state estimate is obtained by the following sum:

$$\text{pdf}(\hat{x}_k^-) = \sum_{i=1}^{M} a_{ki} N(\hat{x}_{ki}^-, P_{ki}^-) \tag{13.105}$$

   (b) The *a posteriori* state estimate is obtained by first executing the following measurement update equations for $i = 1, \cdots, M$:

$$\begin{aligned} H_{ki} &= \left.\frac{\partial h_k}{\partial x_k}\right|_{\hat{x}_{ki}^-} \\ K_{ki} &= P_{ki}^- H_{ki}^T (H_{ki} P_{ki}^- H_{ki}^T + R_k)^{-1} \\ P_{ki}^+ &= P_{ki}^- - K_{ki} H_{ki} P_{ki}^- \\ \hat{x}_{ki}^+ &= \hat{x}_{ki}^- + K_{ki} \left[y_k - h_k(\hat{x}_{ki}^-, 0) \right] \end{aligned} \tag{13.106}$$

   The weighting coefficients a_{ki} for the individual estimates are obtained as follows: